

Pulkit Chawla

Eindhoven, Netherlands | +31 6 45 76 94 65 | pulkitchawl@gmail.com | linkedin.com/in/pulkitchawla11 | github.com/thispc

SUMMARY

Software engineer (6+ years at Oracle Cloud and Qubole) turned AI engineer after an MSc in Data Science and AI at TU Eindhoven. In 2026 designed and shipped four agent systems solo: a gate that decides which AI-written code may ship, a macOS computer-use agent, and an email agent with structural prompt-injection defences. Owns services end to end: design, build, test, release, on-call.

Work authorisation: NL orientation-year permit (full labour-market access); HSM-eligible under the reduced graduate criterion

EXPERIENCE

Freelance and personal projects

Independent AI / software engineer

Jan. 2026 – Present

Eindhoven, Netherlands

- Designed and shipped 4 production agent systems: an AI code verification gate, a macOS computer-use agent, a personal email agent, and a multi-account copy-trading platform.
- Shipped a WhatsApp assistant for an Eindhoven catering business on Twilio, Gemini and FastAPI, handling orders, meal selection and billing questions.

TU/e IMAGEN Programme (with Wageningen University and Hendrix Genetics)

Graduate AI Researcher (MSc thesis), MLOps and Computer Vision

Jan. 2025 – Nov. 2025

Eindhoven, Netherlands

- Built the MLOps platform for IMAGEN, a Dutch government-funded animal-welfare research programme, on SURF's Spider HPC cluster with dCache storage.
- Redesigned a PiCaS pilot-job pipeline into a decoupled control plane: ClearML experiment tracking and a CouchDB job queue feeding SLURM jobs in Singularity containers.
- Raised training throughput to 217.4 images/s from 112.3 at 8 loader workers by converting raw images to HDF5, cutting GPU idle time.
- Exposed training and inference through a web UI and REST API so researchers could fine-tune detection models without logging into the HPC cluster.

Oracle Cloud Infrastructure

Member of Technical Staff, Big Data Service (BDS)

Jan. 2021 – Aug. 2023

Bangalore, India

- Owned the customer-facing OCI Big Data Service console end to end in a 30-engineer team: built, tested, released and operated the Node.js UI.
- Owned the Hive and HBase distributions, forking upstream and back-porting CVE fixes for enterprise customers migrating off Cloudera.
- Served the production on-call rotation, resolving P0 and P1 incidents with US-based support and architecture teams.

Qubole (acquired by Idera)

Software Engineer → Software Engineer II, Cluster Orchestration

Jan. 2018 – Dec. 2020

Bangalore, India

- Owned the cluster-orchestration backend with two other engineers, provisioning Hadoop, Spark, Presto and Hive clusters across AWS, Azure, GCP and OCI.
- Added Datadog cluster monitoring and Jenkins QA pipelines gating every release on client-specific integration tests across staging and production.
- Built secure tunnels between the control plane and clusters inside customer cloud accounts, letting customers operate them from the UI.

PROJECTS & RESEARCH

devteam: a harness of microbots with contracts | *Node.js, JavaScript/TypeScript*

github.com/thispc/devteam

Jul 2026 – present

- Designed a verification gate that agents cannot game: contract hashed into module identity, network-isolated container, held-out tests, and twin runs diffed across two environments.
- Ran an autonomous AI manager that hired its own agent team, drafted contracts from 3 examples and shipped modules through the gate with zero human intervention.
- Modelled unverifiable side effects (network, clock, human input) as declared capability holes with owner-granted allowlists, validated against recorded real traffic.
- Built a headless real-mouse probe layer that caught 15 bugs the unit test suite had passed.

- Built a macOS computer-use agent that perceives UI through the accessibility tree instead of screenshots, keeping all screen data on device.
- Implemented input as real cursor movement along human-like paths and human-speed typing rather than instantaneous programmatic events.
- Kept credentials in the macOS Keychain and exposed only placeholders to the model, substituting secrets at injection time after verifying target site identity.
- Scoped agent actions to an explicit allowlist of permitted applications and domains, blocking interaction with anything outside it.
- Shipped 10 SwiftPM targets in native Swift with every OS primitive behind a protocol boundary, enabling 135 tests to run off-device.

- Contained prompt injection by privilege separation: the model reading raw mail holds no tools, and the acting agent receives only a sanitised summary.
- Gated every outbound message behind one-tap human approval that sends the exact reviewed draft, enforced in code rather than prompt instructions.
- Extended the open-source NanoClaw runtime with an iMessage channel and a portable plain-file knowledge base, adding 16,000 lines.

- Directed and shipped an application pipeline polling 9 applicant-tracking systems across 160+ Dutch employers, matching them against the 12,900-entry IND sponsor register.

- Built a retrieval-augmented question-answering pipeline over 10,000+ academic documents with LangChain, the GPT-4 API and FAISS, holding retrieval under 100 ms.
- Benchmarked retrieval against BM25 and TF-IDF baselines and tuned chunking, context management and prompting to beat plain LLM answers.

EDUCATION

Eindhoven University of Technology (TU/e)	Aug. 2023 – Nov. 2025
<i>Master of Science in Data Science and Artificial Intelligence</i>	<i>Eindhoven, Netherlands</i>
• Thesis: A Reimagined and Implemented MLOps Architecture for Supercomputing Environments (IMAGEN programme, SURF Spider HPC), supervisors Dr. Yanja Dajsuren and Manu Agarwal	
Birla Institute of Technology, Mesra	Aug. 2014 – Jun. 2018
<i>Bachelor of Engineering in Information Technology, CGPA 8.37/10, First Class with Distinction</i>	<i>Ranchi, India</i>

TECHNICAL SKILLS

AI/ML: LLM agents (Claude Agent SDK), RAG and vector search (FAISS), prompt-injection defences, LangChain, PyTorch, computer vision (object detection, tracking), fine-tuning, model serving and inference services, ClearML

Languages: Python, TypeScript/JavaScript, Swift, Java, SQL, Bash

Backend & Tooling: FastAPI, Spring, Node.js, REST APIs, Jenkins, GitHub Actions, GitLab CI, CI/CD, GitOps, Datadog, Grafana, Prometheus, Elasticsearch, Playwright

Big Data & Cloud: Hadoop, Spark, Kafka, AWS, Azure, OCI, GCP, Docker, Kubernetes, Terraform, Ansible, Postgres, Redis

Spoken languages: English: fluent; Hindi: native; Dutch: beginner, learning